

BANK TRANSACTION ANALYTICS DASHBOARD

1. Project Overview

The **Bank Transaction Analytics Dashboard** is a Power BI-based financial analytics project developed to analyze banking transactions and provide a clear overview of transaction activity, deposits, withdrawals, balances, transaction categories, and potential anomalies. The dashboard converts transaction-level data into meaningful KPIs and interactive visualizations that can support financial monitoring and reporting.

2. Project Objectives

- Analyze total banking transactions.
- Monitor deposits and withdrawals.
- Track average account balance.
- Identify monthly transaction patterns.
- Analyze cash flow over time.
- Monitor transaction types.
- Identify the most frequently occurring transaction categories.
- Track balance trends.
- Monitor potentially anomalous transactions.

3. Data Model

Bank_Transaction_Analytics_Final: Main transaction fact table containing banking transaction records.

Calendar: Dedicated date table used for accurate date-based analysis and monthly reporting. The Calendar table supports time-based calculations and improves the reliability of trend analysis.

4. Key Performance Indicators

KPI	Purpose
Total Transactions	Overall number of transactions.
Total Deposit	Total amount deposited.
Total Withdrawal	Total amount withdrawn.
Average Balance	Average account balance.
Total Anomalies	Number of transactions identified as anomalies by the underlying data/model.

5. Dashboard Visualizations

- Monthly Transactions – line chart showing transaction volume across months.
- Monthly Cash Flow – clustered column chart showing the aggregated monthly cash-flow measure.
- Transaction Type Distribution – donut chart displaying transactions by type.
- Monthly Deposit vs Withdrawal – combination chart comparing monthly deposits and withdrawals.
- Top 10 Transaction Categories – clustered bar chart showing the ten most frequently occurring transaction descriptions/categories.

- Balance Trend – line chart showing balance changes over time.

6. Tools and Technologies

- Microsoft Power BI
- Power Query
- DAX
- Data Modeling
- Data Visualization
- Excel/Data Files

7. Business Insights

- Overall transaction volume.
- Changes in deposits and withdrawals.
- Monthly cash-flow patterns.
- Transaction-type composition.
- Frequently occurring transaction categories.
- Balance movement over time.
- Potentially unusual transaction activity.

8. Data Analysis Approach

- Import transaction data into Power BI.
- Clean and transform data using Power Query.
- Create the Calendar table.
- Establish relationships between tables.
- Create DAX measures for important KPIs.
- Build financial and transaction visualizations.
- Organize visuals into a single-page dashboard.
- Validate dashboard results.

9. Conclusion

The Bank Transaction Analytics Dashboard demonstrates how Power BI can transform banking transaction data into an interactive financial monitoring solution. The project combines KPI analysis, time-series analysis, cash-flow monitoring, category analysis, balance tracking, and anomaly monitoring to provide a consolidated view of banking activity.